PHASES OF A COMPILER AND THEIR ROLE

TRANSFORMING SOURCE CODE INTO MACHINE CODE

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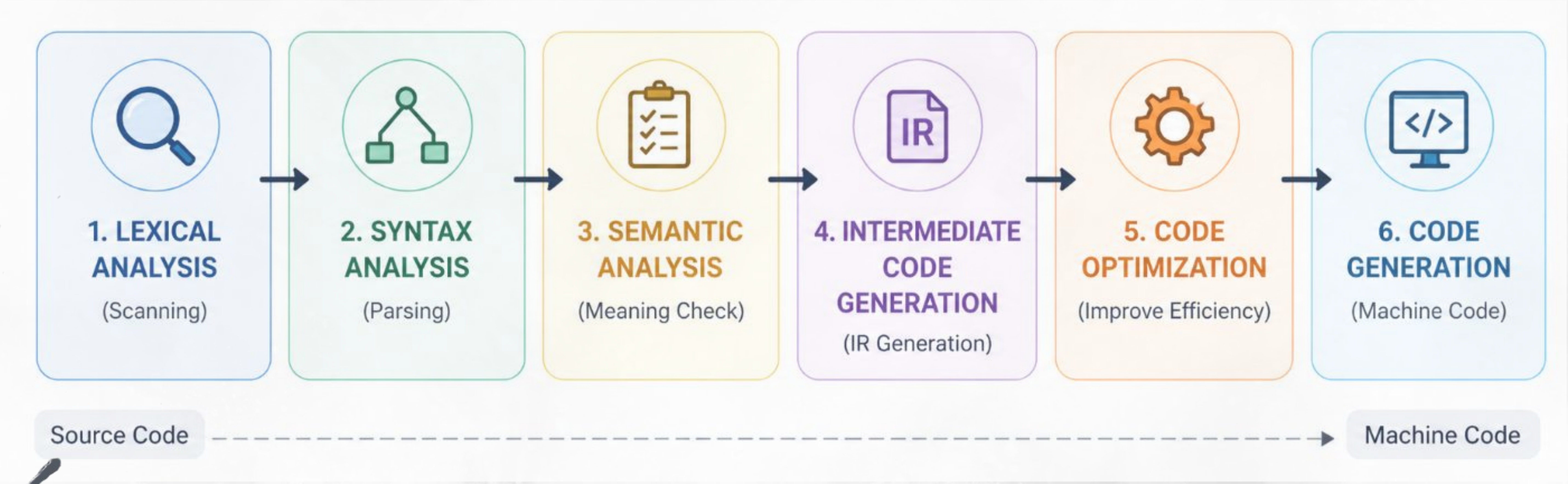
Course Code: CSA1402

Course: Compiler Design

COMPILER PHASES OVERVIEW

Lexical Analysis, Syntax Analysis, Semantic Analysis, Intermediate Code Generation, Code Optimization, Code Generation



ROLE OF EACH PHASE

01 Lexical → converts code to tokens

02 Syntax → checks grammar structure

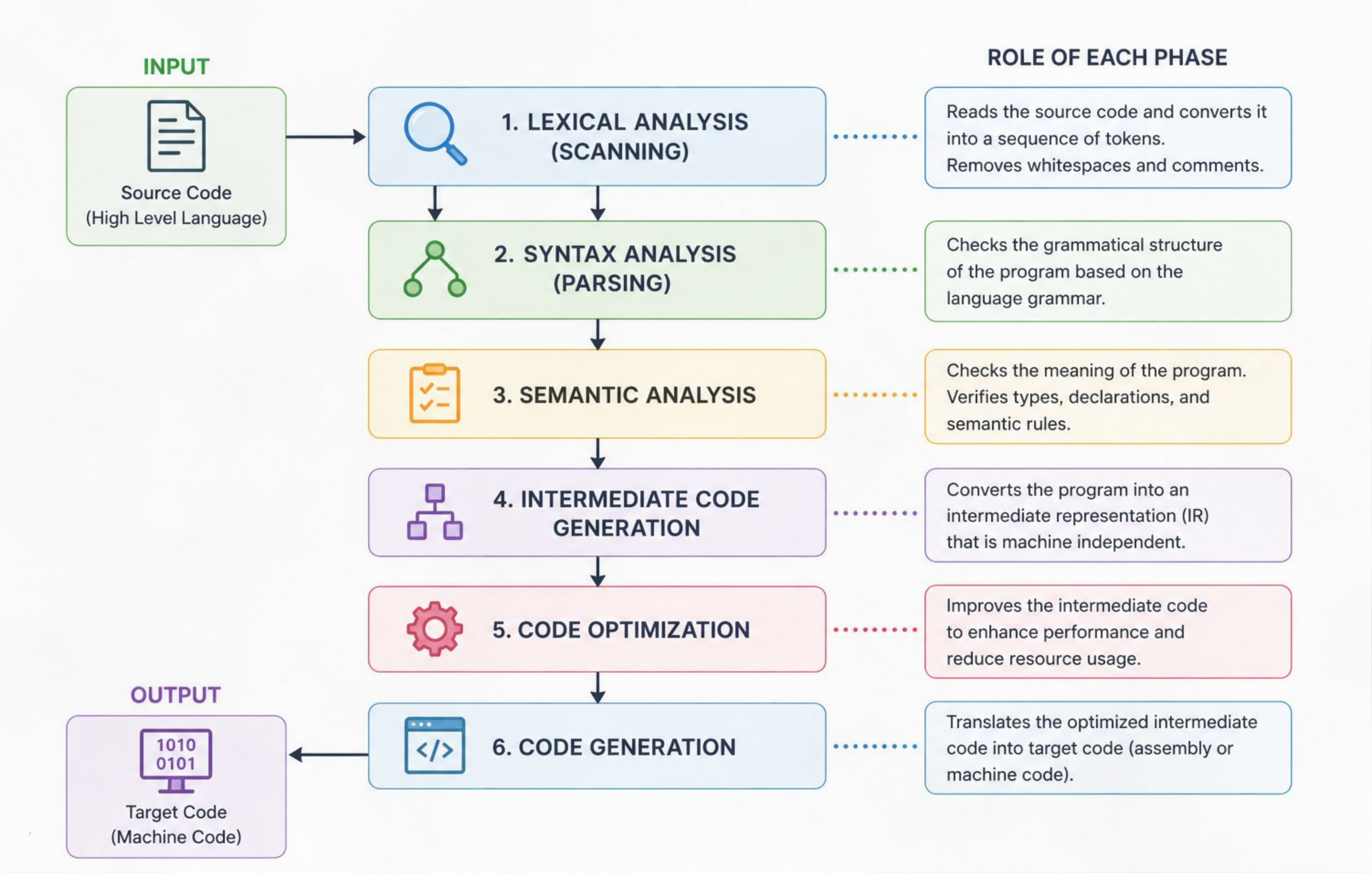
03 Semantic → checks meaning & types

04 Intermediate → creates IR (intermediate code)

05 Optimization → improves performance

06 Code Generation → produces machine code

COMPILER WORKFLOW



ERROR DETECTION BY PHASES

- Lexical → invalid tokens (e.g., @)
- Syntax → missing symbols (e.g., ;)
- Semantic → type mismatch
- Each phase detects specific errors
- Early detection prevents deeper issues

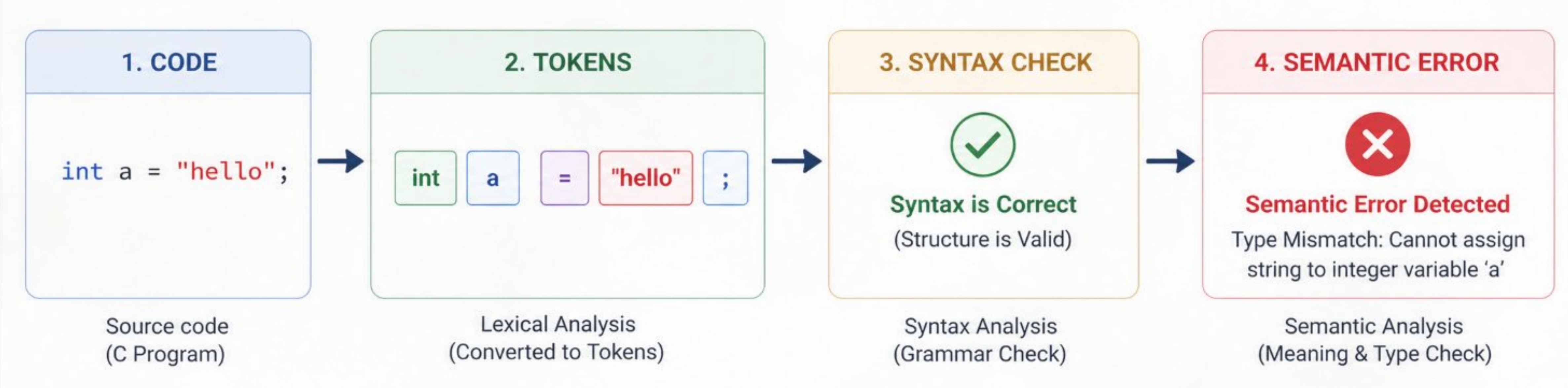
PHASE-WISE ERROR EXAMPLE

CODE :

```c
int a = "hello";
```

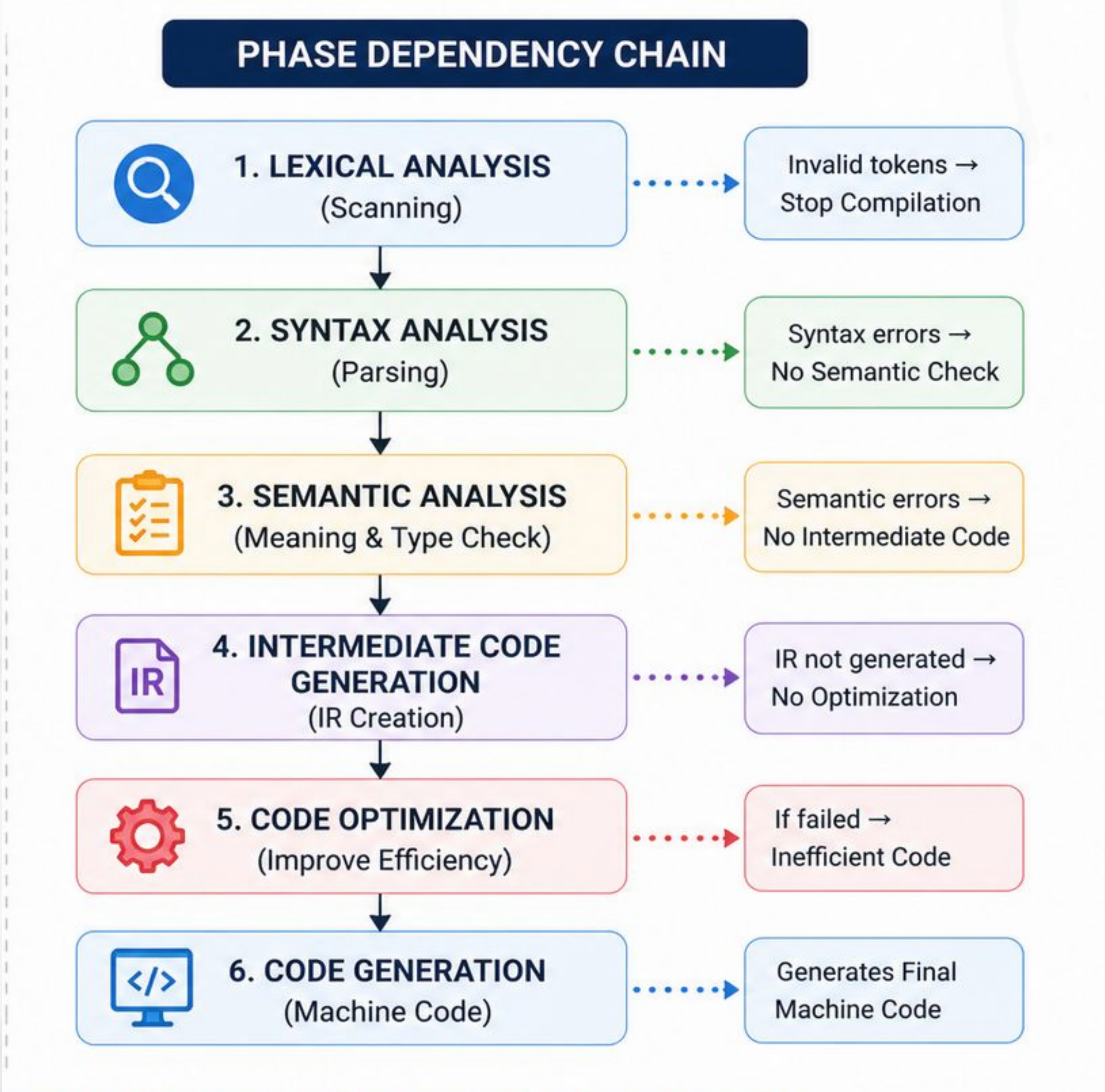
EXPLANATION:

- Lexical → valid tokens
- Syntax → valid structure
- Semantic → type mismatch (int vs string)



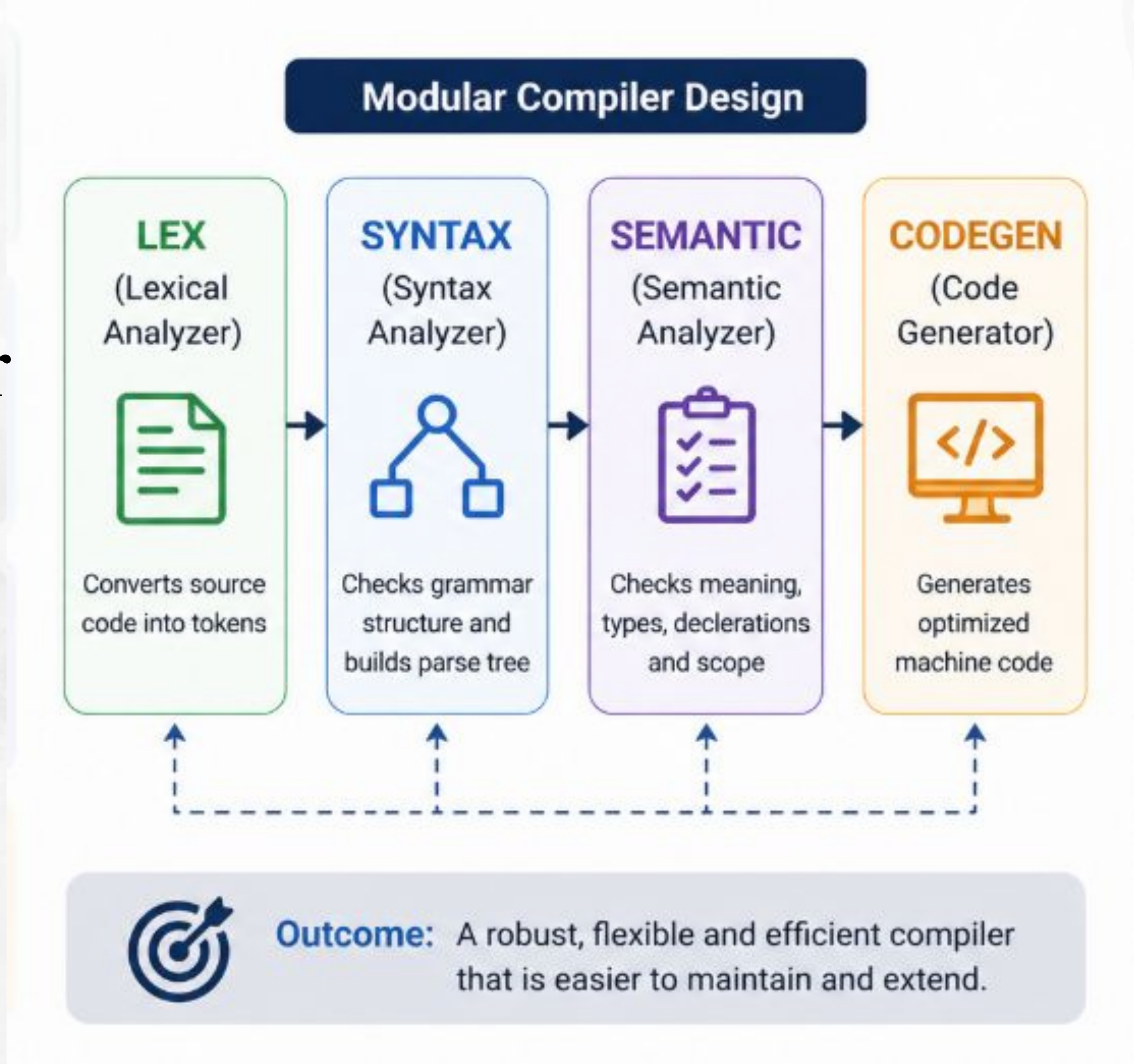
PHASE DEPENDENCY

- Each phase depends on previous output
- Errors in early phase stop execution
- Lexical errors → no parsing
- Syntax errors → no semantic check
- Ensures correctness step-by-step



IMPROVEMENTS IN COMPILER DESIGN

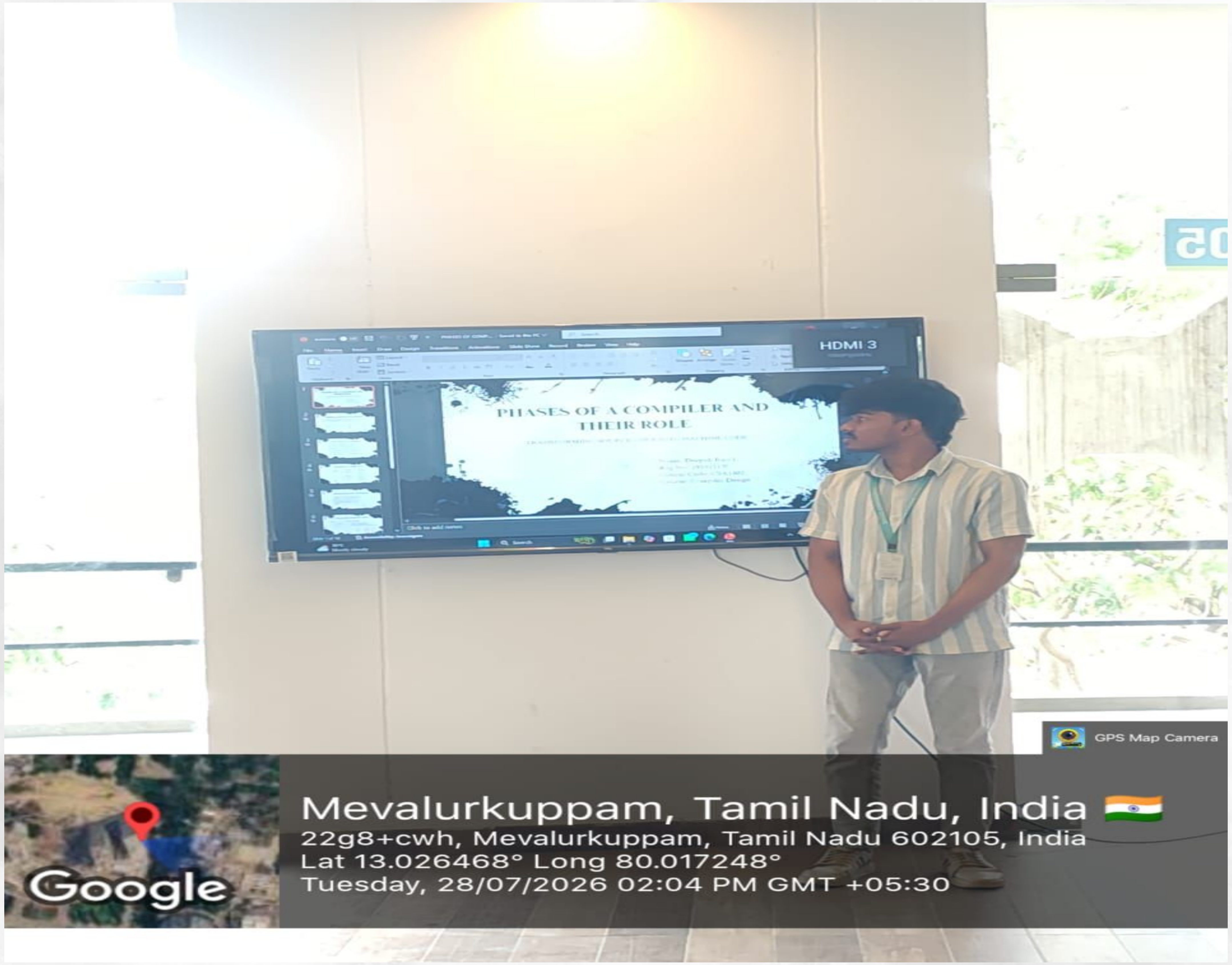
- Add better error recovery techniques
- Allow compilation to continue after minor errors
- Use modular design (independent phases)
- Improves debugging and flexibility



CONCLUSION

Compiler phases convert source code into machine code through a structured, step-by-step process, where each phase has a clearly defined role. Early error detection ensures efficiency by stopping incorrect programs at the right stage. The dependency between phases guarantees accuracy, while a modular design improves flexibility, debugging, and overall performance.

THANK YOU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